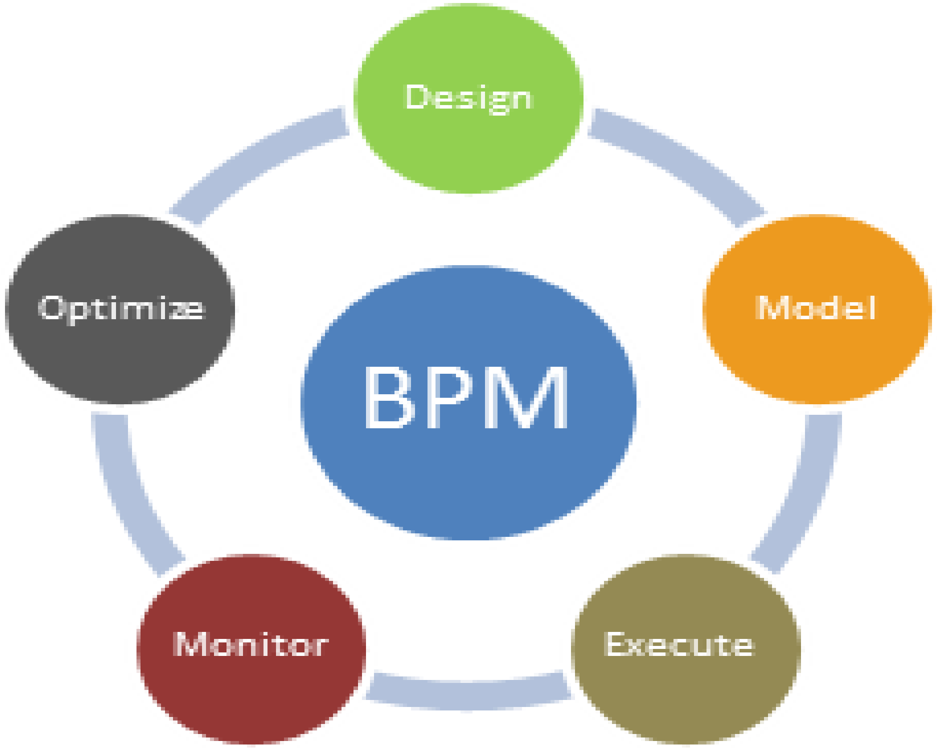
CSE345
Lect 2

Business Process Management



Dr. Islam El-Maddah

BPMN

- Business Process Modeling Notation
- Inspired by cross-functional flowcharts
- OMG Standard, supported by many tools:
  - TIBCO Business Studio
  - IBM Websphere Business Modeler
  - ARIS, Oracle BPA
  - Telelogic System Architect (now IBM)
  - ITP Commerce Process Modeler for Visio
  - Oryx (http://bpt.hpi.uni-potsdam.de/Oryx)
  - Savvion, Lombardi, BizAgi, ...
  - https://bpmn.io/

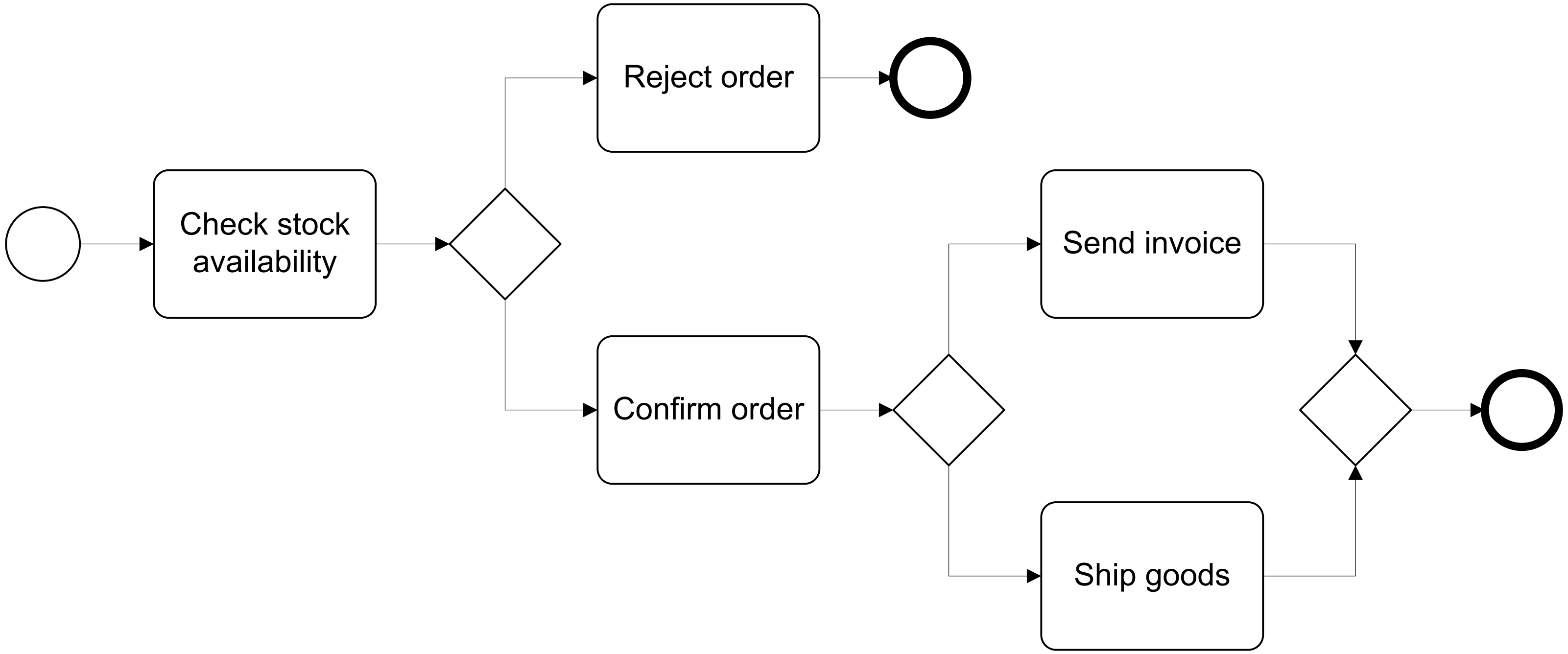
BPMN from 10 000 miles…

- A process model in BPMN is called a Business Process Diagram (BPD)
- A BPD is essentially a graph consisting of four types of elements (among others):

Example

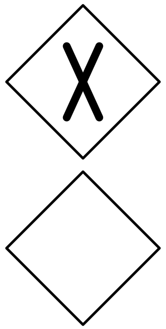
An Order Management process is triggered by the reception of a purchase order from a customer. The purchase order has to be checked against the stock. Depending on stock availability the purchase order may be confirmed or rejected. If the purchase order is confirmed, the goods requested are shipped and an invoice is sent to the customer.

Order Management Process in BPMN

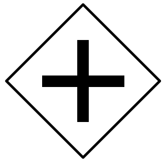


A little bit more on gateways …

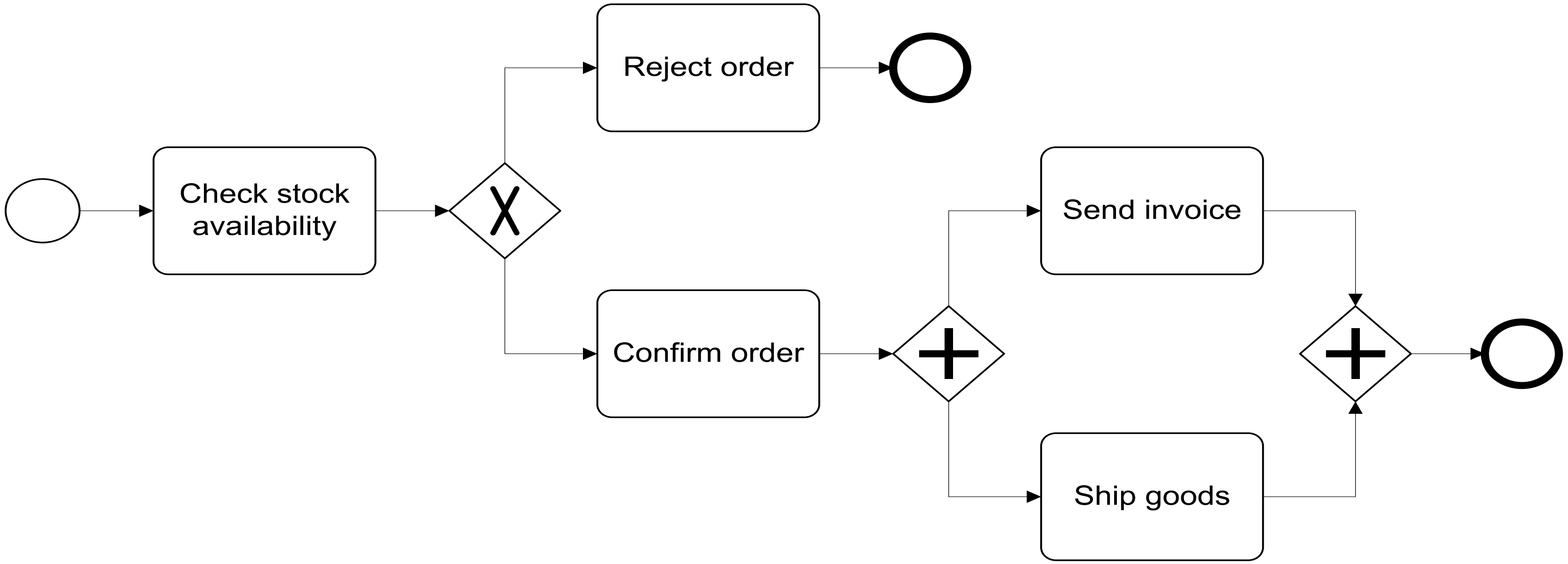
- Exclusive Decision / Merge
  - Indicates locations within a business process where the sequence flow can take two or more alternative paths.
  - **Only one** of the paths can be taken.
  - Depicted by a diamond shape that *may* contain a marker that is shaped like an “X”.



- Parallel Fork / Join
  - Provide a mechanism to synchronize parallel flow and to create parallel flow.
  - Depicted by a diamond shape that *must* contain a marker that is shaped like a plus sign.



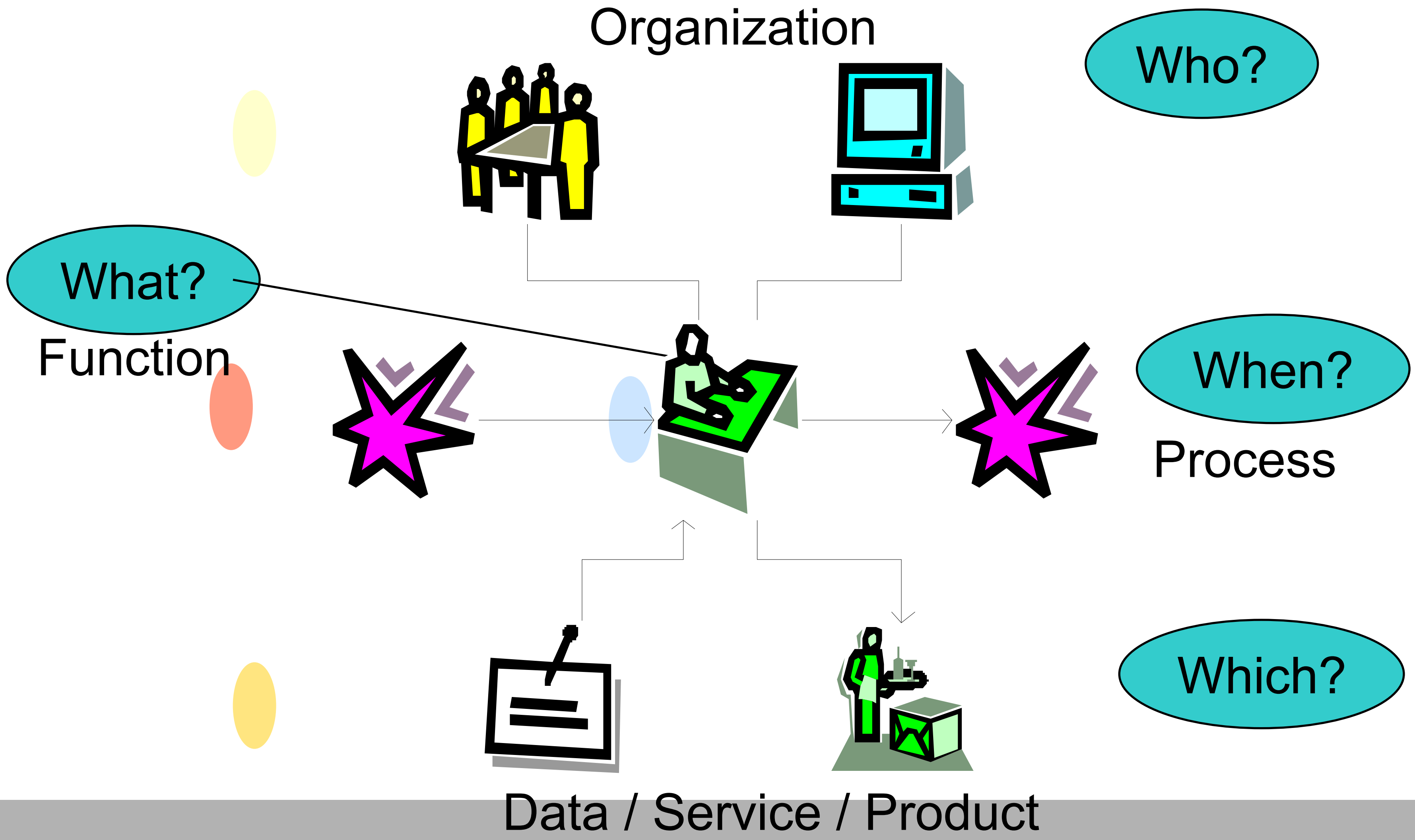
Revised Order Management Process



BPMN Exercise 1:
Claims Notification process at a car insurer

When a claim is received, it is first checked whether the claimant is insured by the organization. If not, the claimant is informed that the claim must be rejected. Otherwise, the severity of the claim is evaluated. Based on the outcome (simple or complex claims), relevant forms are sent to the claimant. Once the forms are returned, they are checked for completeness. If the forms provide all relevant details, the claim is registered in the Claims Management system, which ends the Claims Notification process. Otherwise, the claimant is informed to update the forms. Upon reception of the updated forms, they are checked again.

Process Modelling Viewpoints



Process Modelling Viewpoints

Functional perspective

What tasks/function are happening in the process?

Control-flow perspective

In what order do they occur?

Resource perspective (also called organisational perspective)

Who performs which activity?

Data perspective

What data are created/produced by the process?

Organisational Elements in Process Models

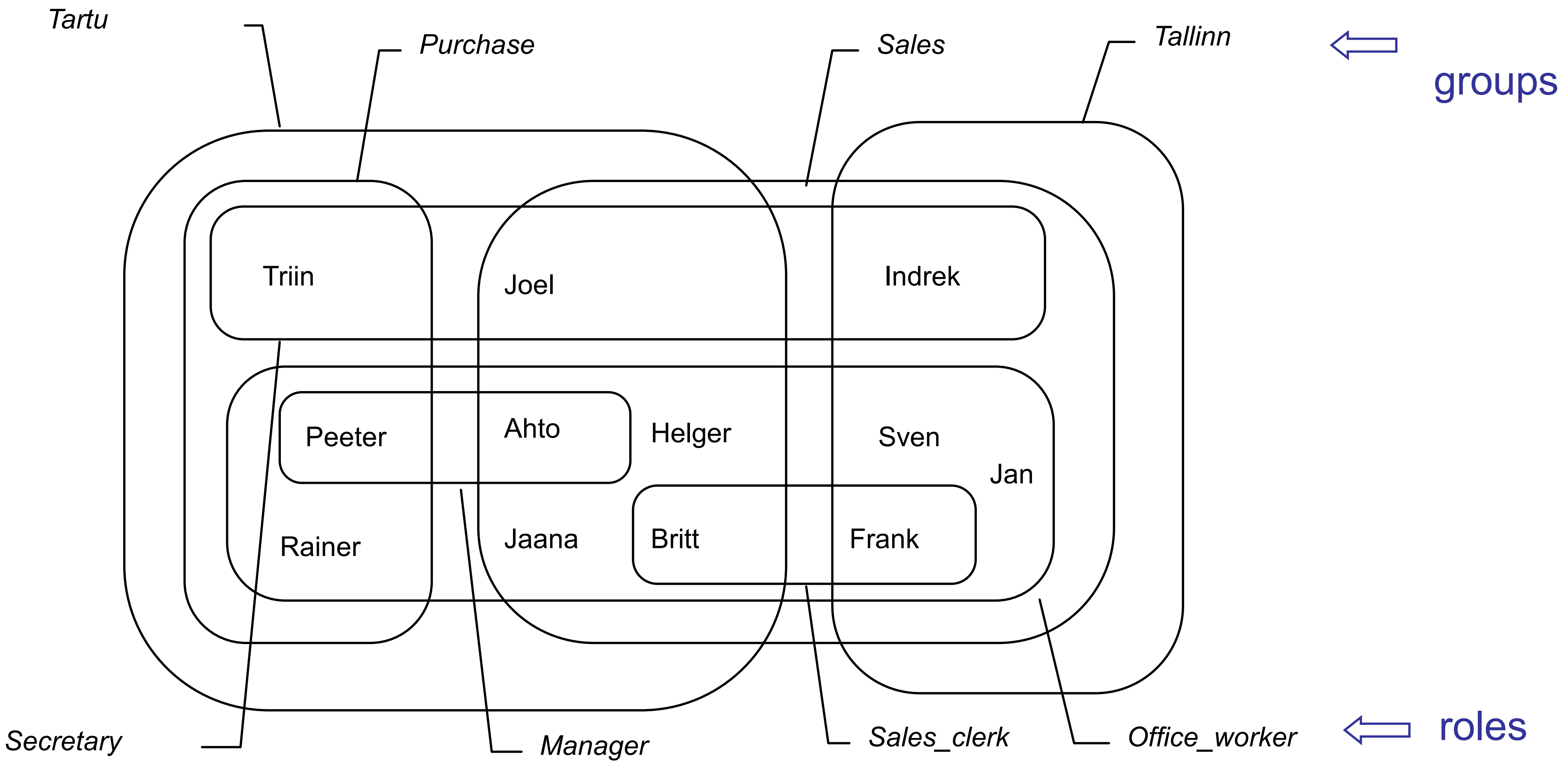
Basic abstractions:

- **Resource** (participant, actor, user, agent)
  A resource can execute certain tasks for certain cases.
  Human and/or non-human (e.g. printer).
- **Resource class:** Set of resources with similar characteristics

A *resource class* is typically either a:

- **Role** (skill, competence, qualification)
  Classification based on what a resource can do or is expected to do
- **Group** (department, team, office, organizational unit)
  Classification based on the organization.

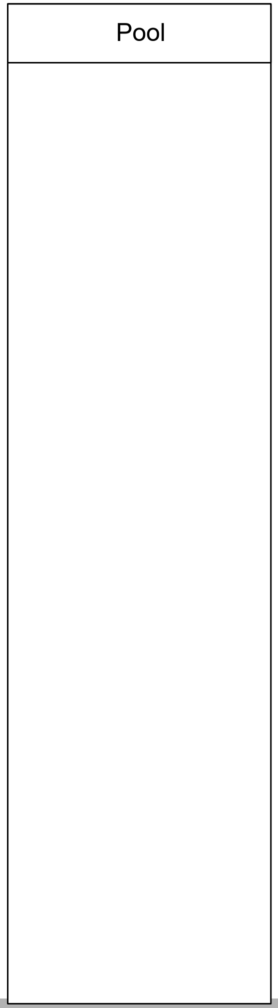
Roles vs. Groups



Resource Modelling in BPMN

- In BPMN, resource classes are captured using:
  - Pools – independent organisations or organizational units (e.g. customer, supplier, East-Tallinn Hospital, Tartu Clinic)
  - Lanes – tightly connected roles or groups (e.g.

BPMN Elements – *Pools*



- *Pools* represent business process participants. They are used to partition a set of activities.
  - Can be a business *entity* or a business *role.*

- Sequence flows cannot cross the boundaries of a Pool.

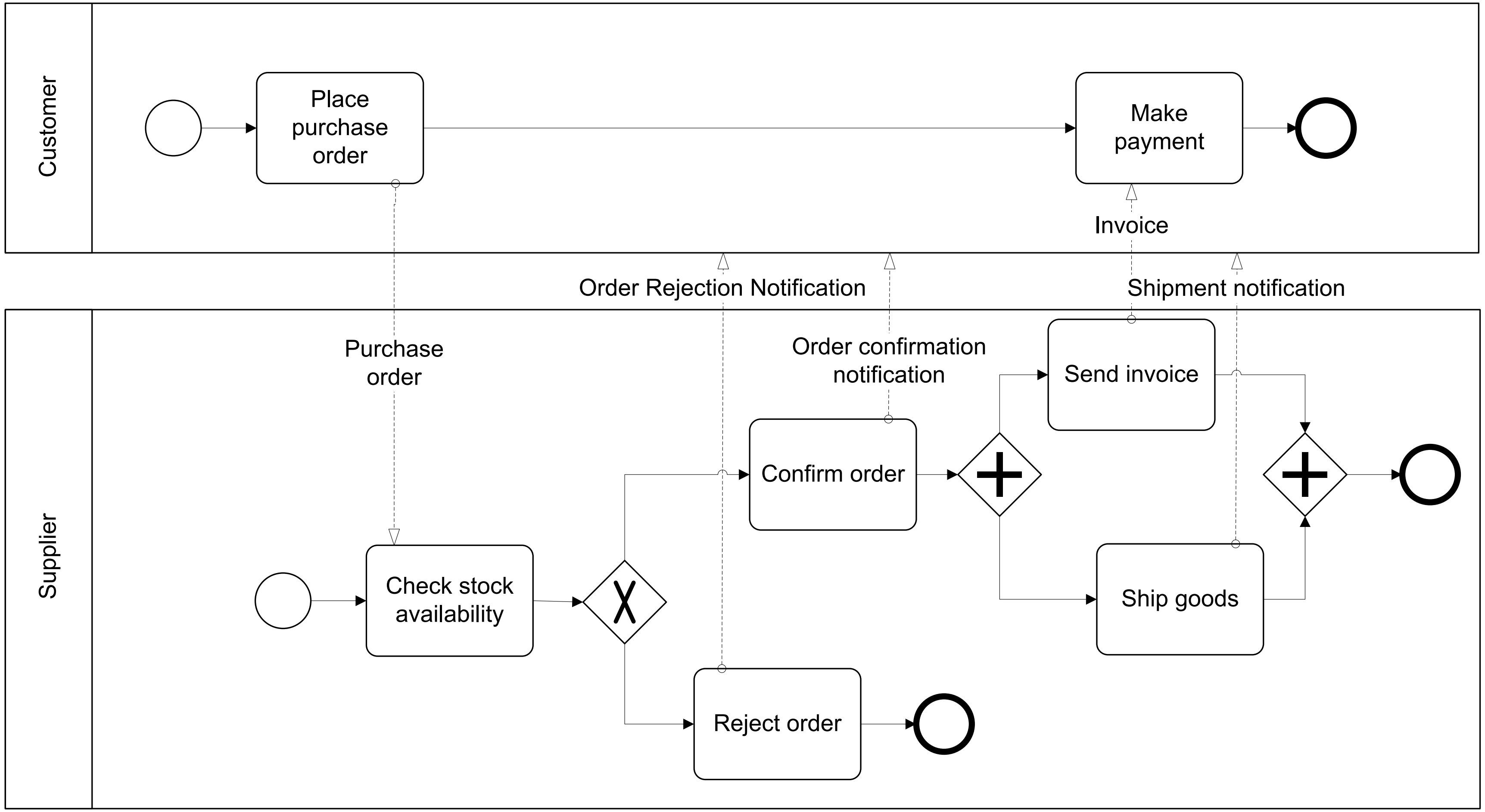
- Interaction between Pools are captured through *Message Flow (dashed lines with an arrow)*

Order Management example (ctd.)

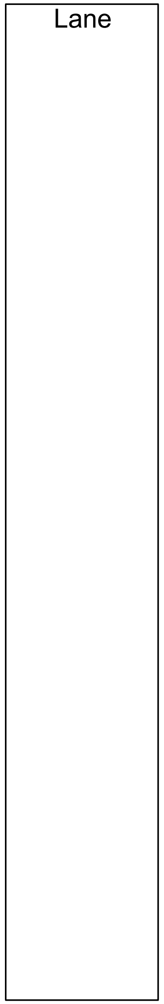
- The Order Management process now includes the customer as a process participant...

  The Order Management process is started when a customer places a purchase order. The purchase order has to be checked against the stock re the availability of the product(s). Depending on stock availability the purchase order may be confirmed or rejected. If the purchase order is confirmed, the goods requested are shipped and an invoice is sent to the customer. The customer makes then makes the payment.

Order Management BPD with Swimlanes



BPMN Elements – *Swimlanes*



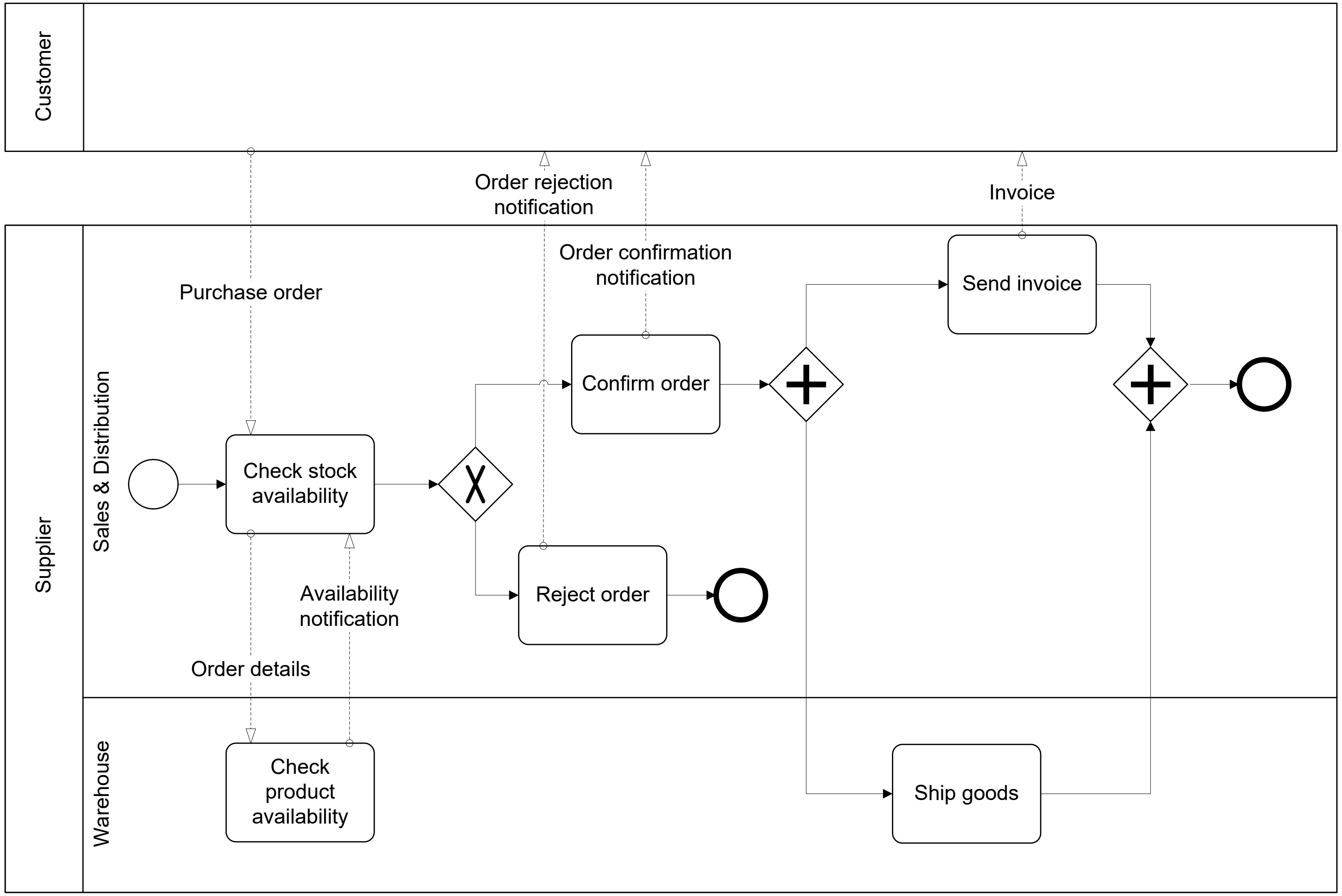
- *Lanes* represent sub-partitions *within* a pool. They are used to organize and categorize activities.
  - Horizontal vs. vertical
  - Meaning is not specified by BPMN, Lanes are often used for internal roles (e.g., Manager, Associate), systems (e.g., an enterprise application), an internal department (e.g., shipping, finance), etc.
- Both Sequences Flow and Message Flow can cross the boundaries of Lanes.
- Lanes can be nested:
  - E.g., an outer set of Lanes for company departments and then an inner set of Lanes for roles within each department.

Order Management example (ctd.)

- The process now includes two departments within the supplier organization...

  The purchase order received by the Sales & Distribution department has to be checked against the stock. The order details are sent to the Warehouse department that returns an availability notification. If the purchase order is confirmed, the Warehouse department collects the shipping details from the customer and ships the goods. The Sales & Distribution department sends an invoice to the customer who then makes the payment.

Corresponding BPMN Model



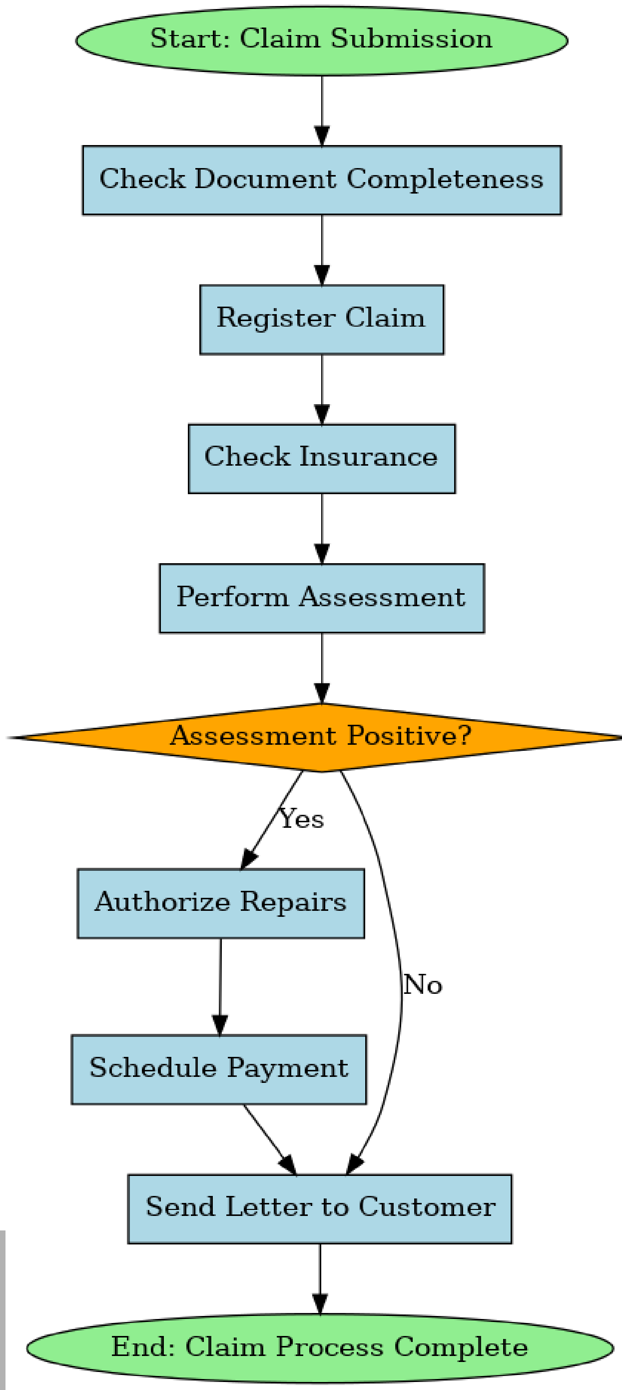
BPMN Exercise 2: Lanes, Pools

- Claims Handling process at a car insurer

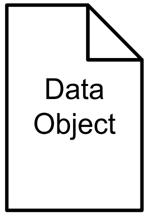
  A customer submits a claim by sending in relevant documentation. The Notification department checks the documents for completeness and registers the claim. The Handling department picks up the claim and checks the insurance. Then, an assessment is performed. If the assessment is positive, a garage is phoned to authorise the repairs and the payment is scheduled (in this order). In any case (whether the outcome is positive or negative), a letter is sent to the customer and the process is considered to be complete.

Solutions

1. **Start Event**: "Claim Submission" (Customer submits a claim)
2. **Task**: "Check Document Completeness" (Notification department checks documents)
3. **Task**: "Register Claim" (Notification department registers the claim)
4. **Task**: "Check Insurance" (Handling department reviews insurance details)
5. **Task**: "Perform Assessment" (Assessment is conducted)
6. **Gateway**: "Assessment Positive?" (Decision point)
    1. **Yes Path**:
        1. **Task**: "Authorize Repairs" (Garage is contacted for repair authorization)
        2. **Task**: "Schedule Payment" (Payment is arranged)
    2. **No Path**: Skip repair and payment.
7. **Task**: "Send Letter to Customer" (Notify customer of the outcome)
8. **End Event**: "Claim Process Complete"

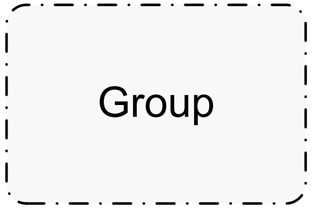


BPMN *Artifacts*



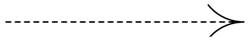
- *Data Objects* are a mechanism to show how data is required or produced by activities.
  - Represent input and output of a process activity.

- *Annotations* are a mechanism for the modeller to provide additional text information to the diagram reader.
  - Text annotations do not affect the flow of the process.



- *Groups* are a visual mechanism to logically group diagram elements informally.

BPMN *Connections*



Directed association

Undirected association

- *Associations* are used to link artifacts such as text or data objects with flow objects.
  - Are depicted by a dotted line.
  - Can be directed or undirected.
- They can be used to show inputs and outputs of activities.

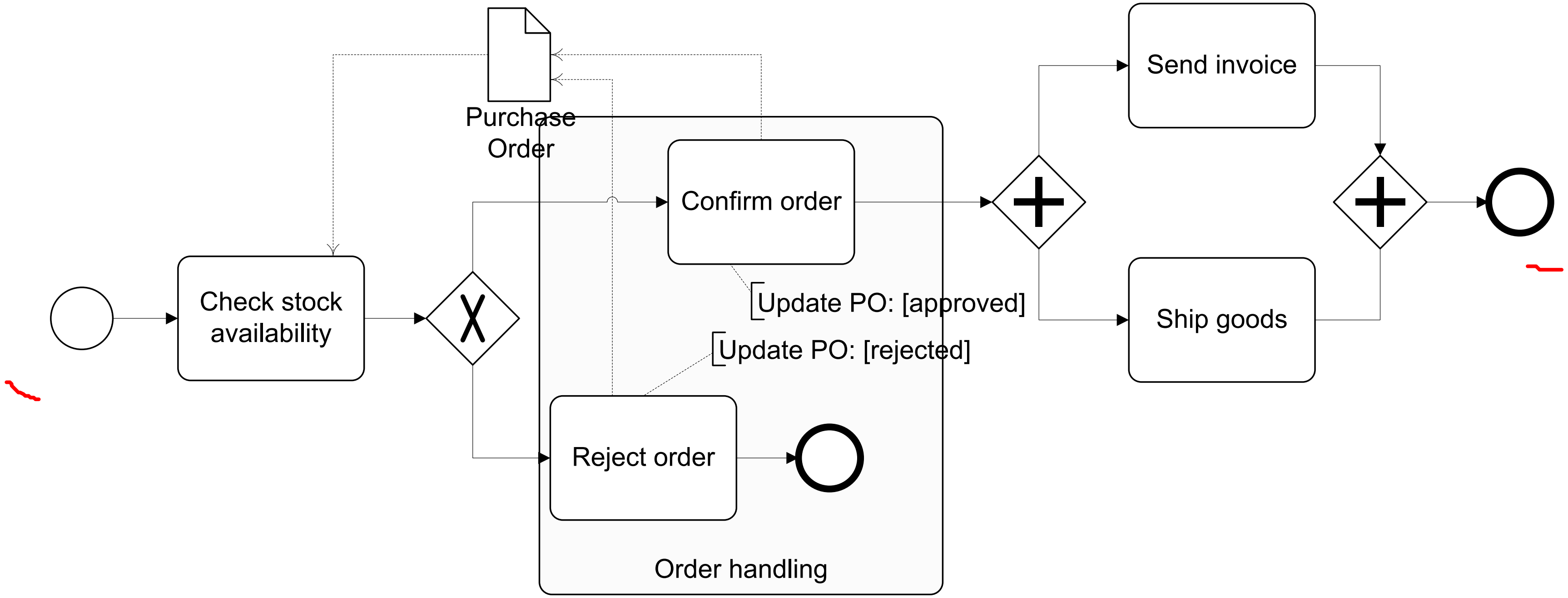
Order Management example (ctd.)

- Let’s have a look at some artifacts...

  The Purchase Order document serves as an input to the stock availability check. Based on the outcome of this check, the status of document is updated, either to “approved” or “rejected”.

  Include the relevant documents in the process model. Also, for visualization purposes, all parts of the processes that use or update the purchase order should be highlighted.

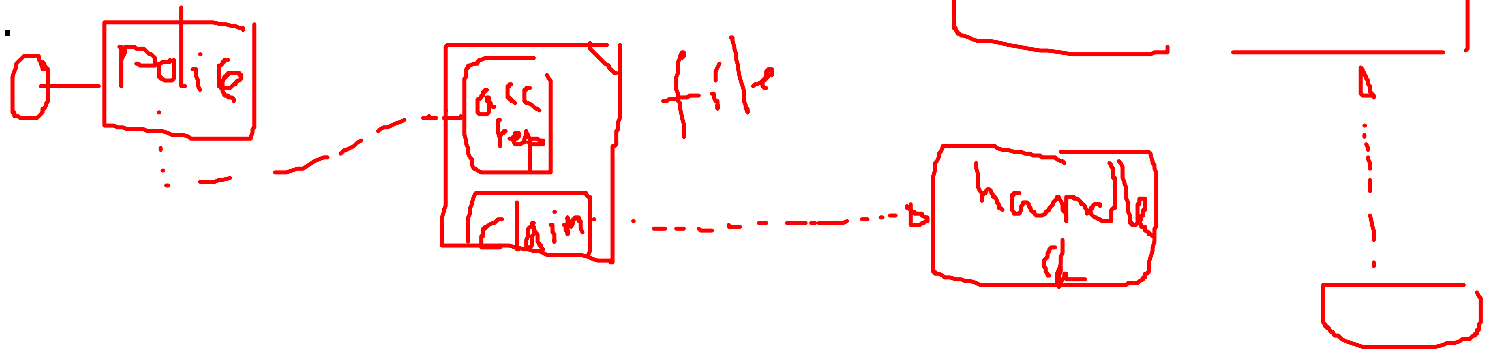
Order Processing Model with Artifacts



BPMN Exercise 3: Artifacts

The report related to the car accident is searched within the Police Report database and put in a file together with the claim documentation. This file serves as input to a claims handler who calculates an initial claim estimate. Then, an action plan is created based on a checklist available in the Document Management system. Based on the action plan, a claims manager tries to negotiate a settlement on the claim estimate. The claimant is informed of the outcome, which ends the process.

Please depict all relevant documents in the model. Also visualize activities that are performed by the claims handler.



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